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C1202 USB Datenanfrage Analyse

1. Messaufbau:

1.1. C1202 mit FW 1.1.0.0

1.2. Digimatic Kabel: DK-U1

1.3. Messung: Saleae logic Pro 8

1.4. Empfänger/Anförderung: PC06-451

2. Interface Beschreibung

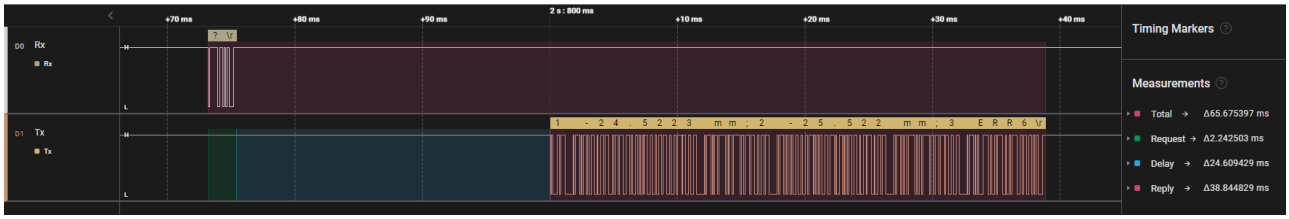
- interfcae: 9600 7E2
  20220926_DK-U1 Instruction Set.xlsm

> **\essfs01\Abteilungen\Benutzergruppen\Firmware_Software\Interface Beschreibung Datenkabel DK-U1\Release\20220926_DK-U1 Instruction Set.xlsm**

3. Messungen:

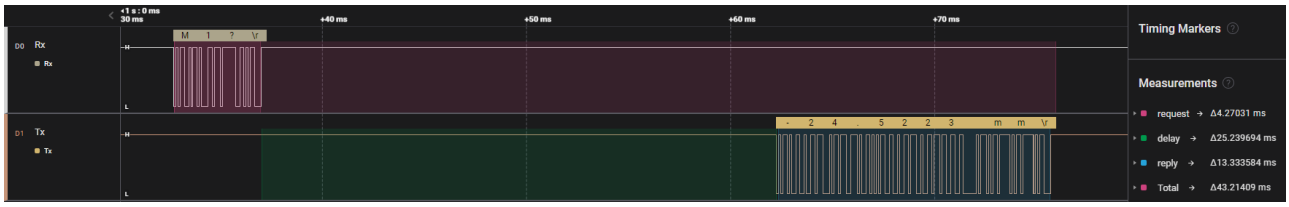
3.1. Einzelantwort für '?':

- Anzahl Merkmale: 1, 2 oder 3
- Toleranz: Nein
- Warngrenzen: Nein
- Anforderung: Befehl ('?')
- Daten Senden: Parallel
- Gesamtzeit: 60ms



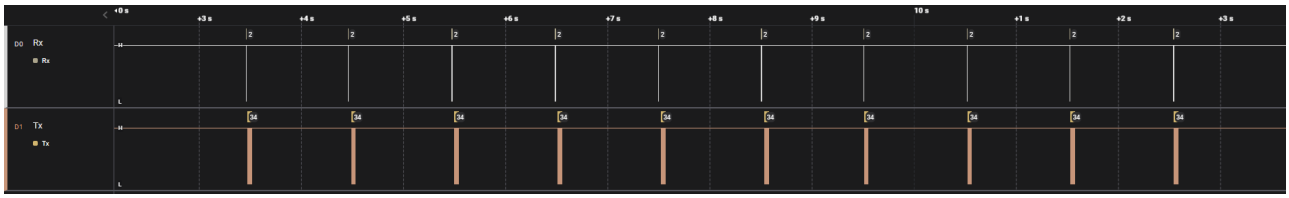
3.2. Einzelantwort für 'M1?':

- Anzahl Merkmale: 1, 2 oder 3
- Toleranz: Nein
- Warngrenzen: Nein
- Anforderung: Befehl ('M1?')
- Daten Senden: Parallel
- Gesamtzeit: 40ms

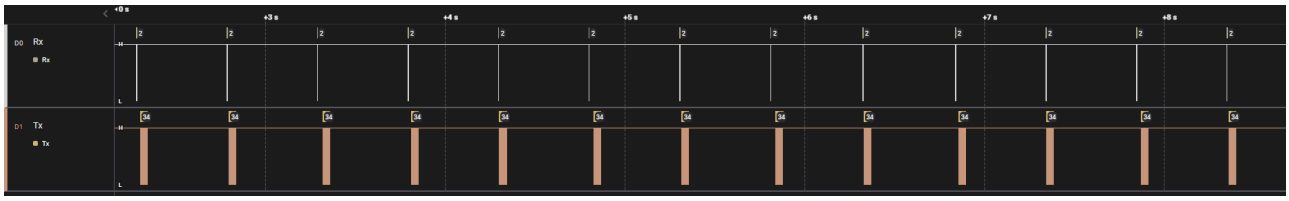


3.3. Zyklischeanforderung für alle Merkmale:

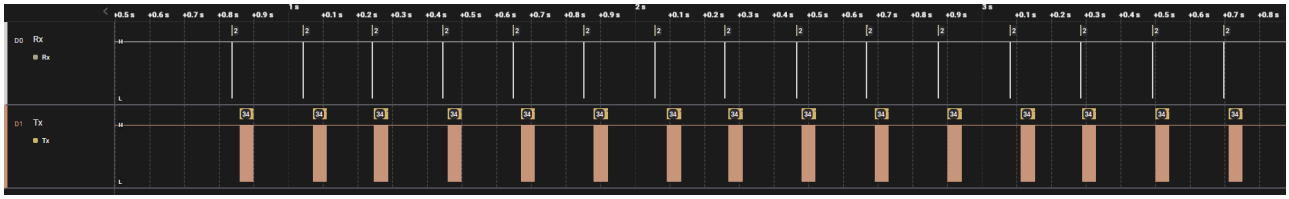
- 1000ms



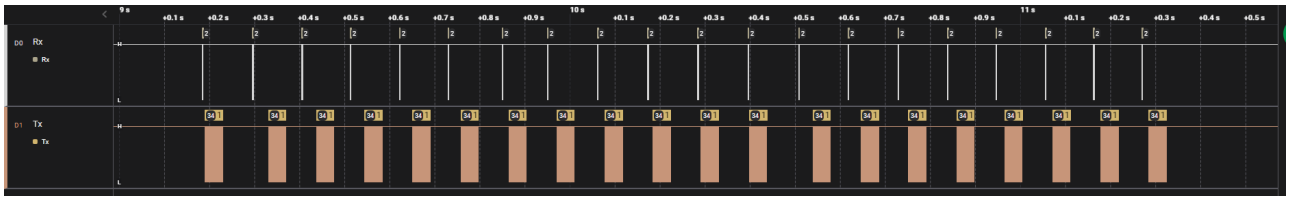
- 500ms



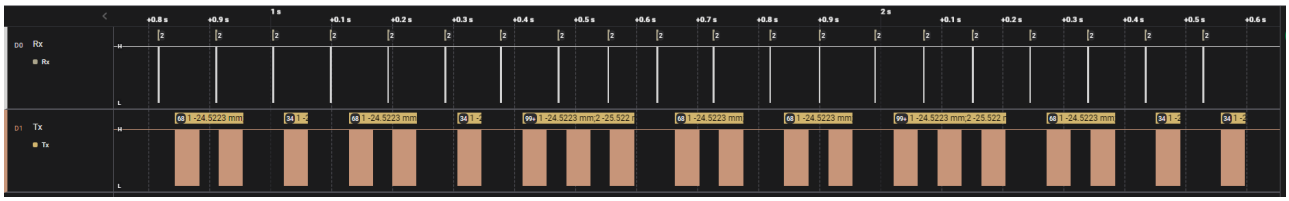
- 200ms



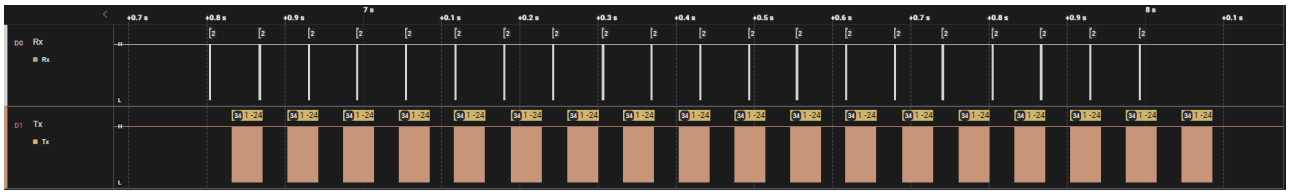
- 100ms

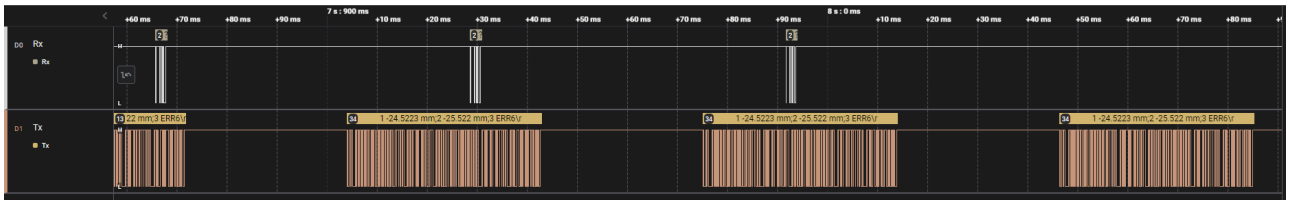


- 80ms

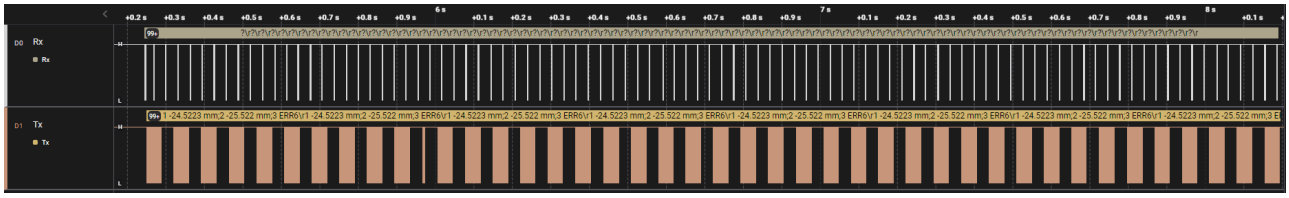


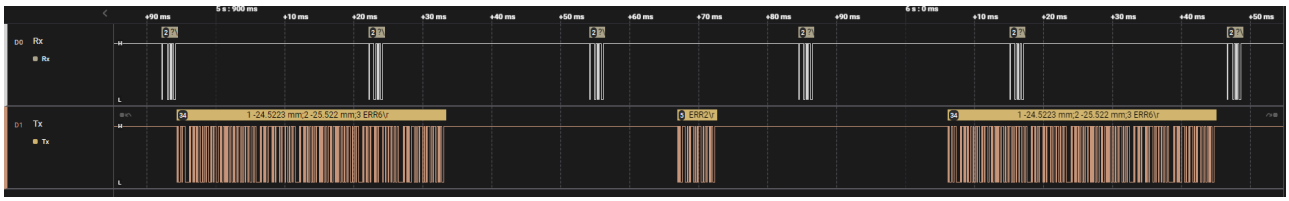
- 50ms



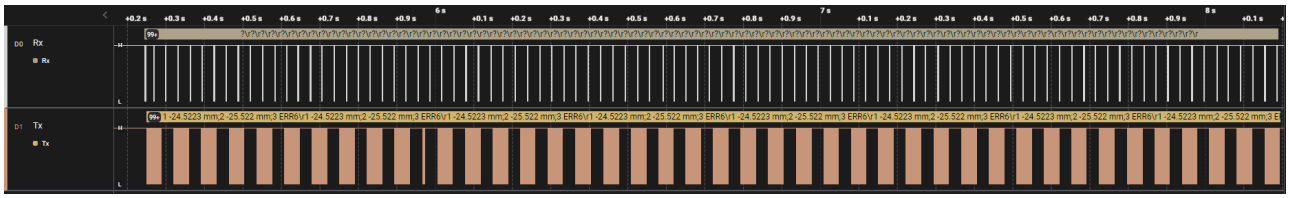


- 20ms



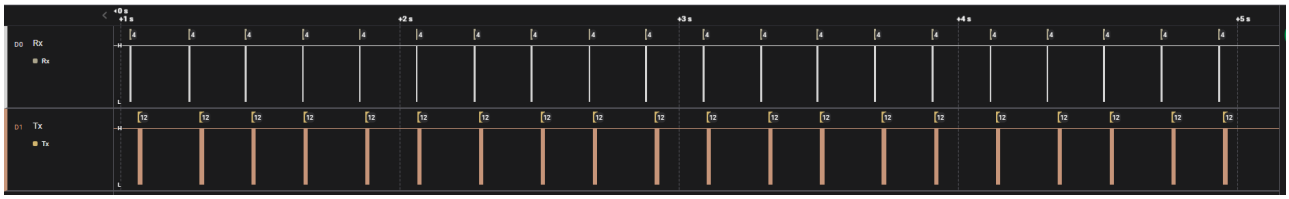


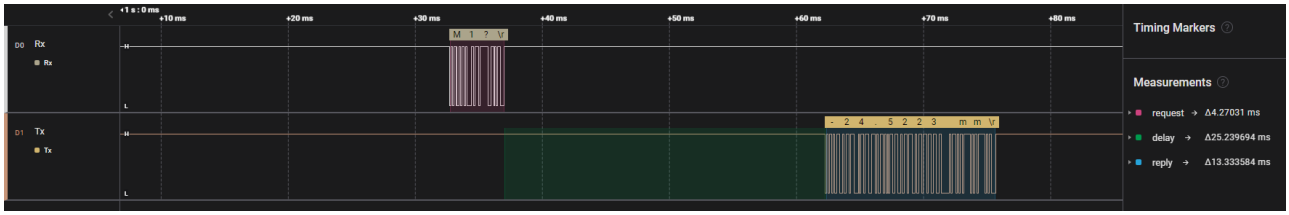
- 10ms



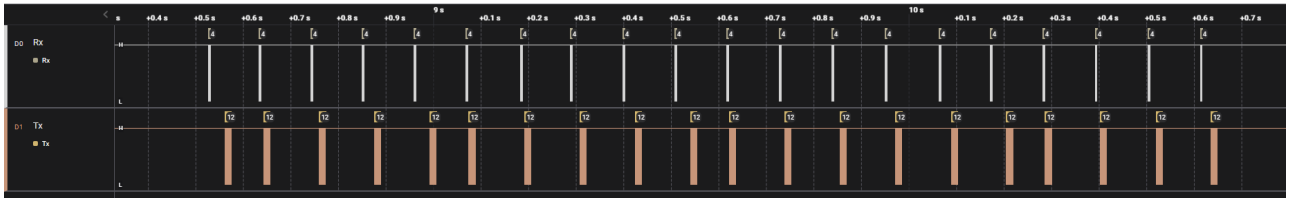
3.4. Zyklischeanforderung für Einzelmerkmal: ('M1')

- 200ms:

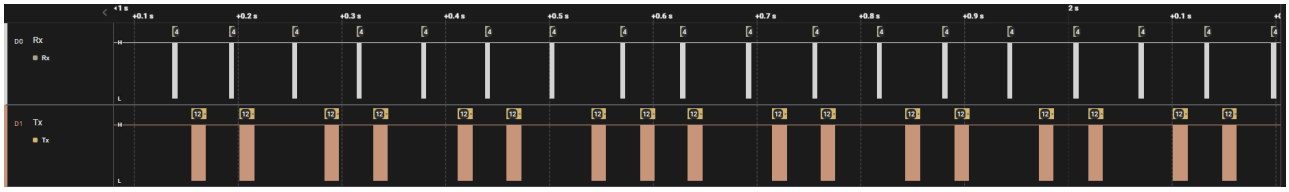




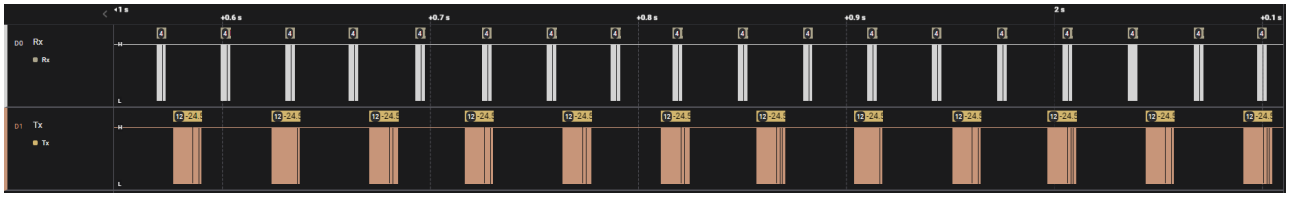
- 100ms:



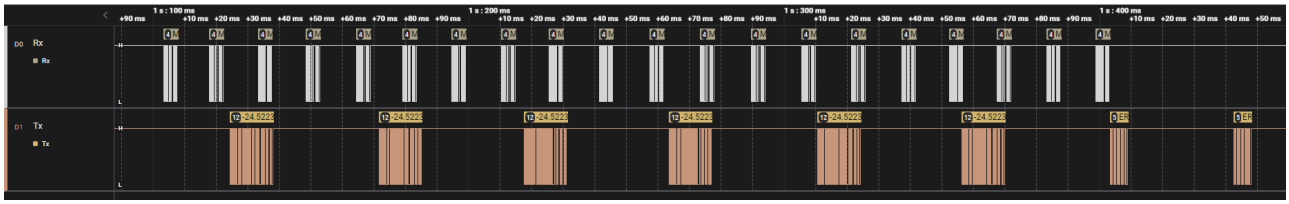
- 50ms:



- 20ms:

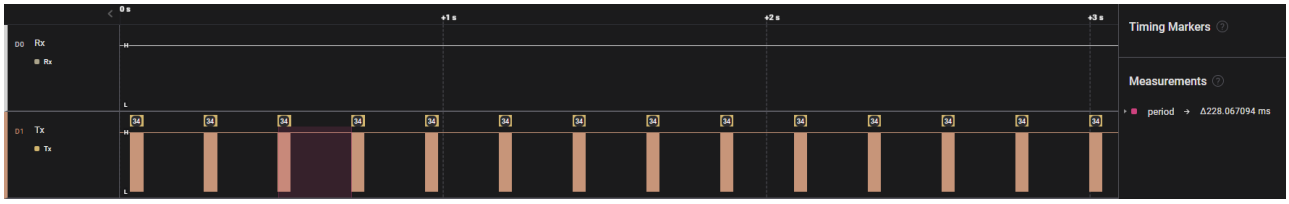


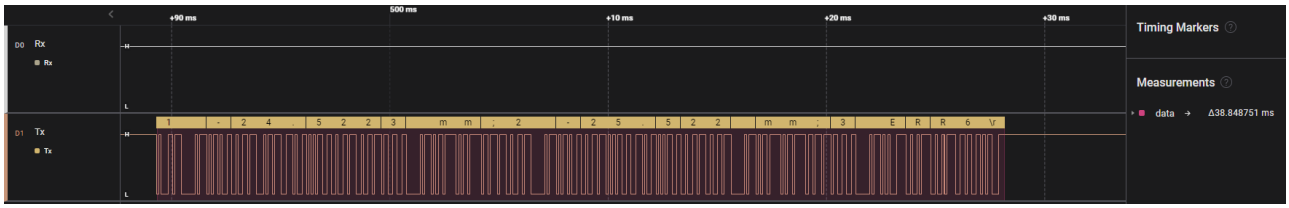
- 10ms:



3.5. Automatische senden für Zeit gesteuerte Messung

- Schnellste Einstellung ist: 200ms (Messdauer=0,1 sek, Pausendauer=0,1 sek):





Nachrichtlänge: ca. 40ms

4. Ergebnis:

- Ausgabewerte sind plausiebel mit aktuellem Werten.
- Limietireungsfaktor ist baud rate: 9600 bps
- Zeiten:
    - Kurze Anfrage ('?'): ca 2ms
    - Länge Anfrage ('Mx?'): ca 4ms
    - Antwort verzügerung: ca 20ms
    - Nachricht ca 15-40ms
    - Frage-Antwort: 40-60ms
    - Maxmimale Anforederungsrate: 100ms (10 Werte pro sekunde)

5. Zeitverkürzungs möglichkeiten (Mehr Data Anfordern):

- Einzelwert Anfragen: 'Mx?'
- Baudrate erhühren, (Antwortzeit ca. 30ms)
- digimatic Schnittstelle nutzen

Source code for testing

> Python / Jupyter Notebook

```
import serial
import time
import ipywidgets as widgets

dataReq = '?\r'.encode('ASCII')

ser = serial.Serial()
ser.baudrate = 9600
ser.port = 'COM5'
ser.bytesize = 7
ser.parity = 'E'
ser.stopbits = 2
ser.timeout = 0

def runtest(time_ms, request_qty):
    ser.open()
    print("start: " + str(time_ms/1000) + " s ...")
    for i in range(request_qty):
        ser.write(dataReq)
        time.sleep(time_ms/1000)
    print("finished")
    ser.close()
widgets.interact_manual(runtest, time_ms=widgets.FloatSlider(min=10, max=1000,
step=10), request_qty=widgets.IntSlider(min=10, max=100, step=10));
```